

Monetary Trends



Discounting the Discount Rate

During the first eleven months of 2001, the Federal Open Market Committee (FOMC) announced ten reductions in its target for the federal funds rate. Each of the announcements also included a statement about the discount rate. For example, the November 6 announcement that the federal funds rate target was being lowered by 50 basis points to 2 percent also noted that “In a related action, the Board of Governors approved a 50 basis point reduction in the discount rate to 1-1/2 percent.” In fact, changes in the discount rate have been identical in timing and magnitude to changes in the federal funds rate target since mid-1999.

The accompanying figure shows that the perfect one-for-one movement between these two key interest rates is a fairly recent development. During the late 1980s and early 1990s, the federal funds rate target deviated from the discount rate by varying magnitudes, with the discount rate changing less frequently than the intended federal funds rate. Nevertheless, the announcements accompanying discount rate changes often explained that changes were made simply to keep the discount rate aligned with market interest rates.

The discount rate has not always had such a passive role in the conduct of monetary policy. During the 1960s and 1970s, for example, the discount rate often served a signaling function: Changes in the FOMC’s objective for the federal funds rate were not publicly announced at the time, and the visibility of discount rate changes made it a useful tool for communicating major changes in policy. During the years when the Fed used a nonborrowed reserve operating target for conducting monetary policy (1979-82), the spread between the federal funds rate and the discount rate was highly variable as changes in reserve demand were accommodated by fluctuations in discount window borrowing. That operating regime was succeeded by one in which the Fed ostensibly targeted the amount of bor-

rowed reserves, amounting to a procedure for manipulating the spread between the funds rate and discount rate. Under the current operating procedure, the federal funds rate is the direct instrument of policy, with the discount rate and the very low average level of discount borrowing having little or no significance in the conduct of monetary policy.

Given the reduced importance of the discount rate in the Fed’s conduct of monetary policy, there have been proposals to dramatically alter the nature of the discount window. Some have suggested that the discount rate be adjusted automatically with changes in the federal funds rate or other market rates. Other proposals have even suggested the elimination of the discount window altogether.

The surge in borrowing following the September 11 terrorist attacks demonstrated, however, that the discount window can perform a crucial stabilizing function, providing temporary liquidity in times of crisis. Moreover, the evolution of the Federal Reserve’s operating procedures has shown that the importance of policy tools can change over time. While the discount rate has little independent significance in the present context, it may again sometime in the future.

—Michael R. Pakko

The Intended Federal Funds Rate and the Discount Rate

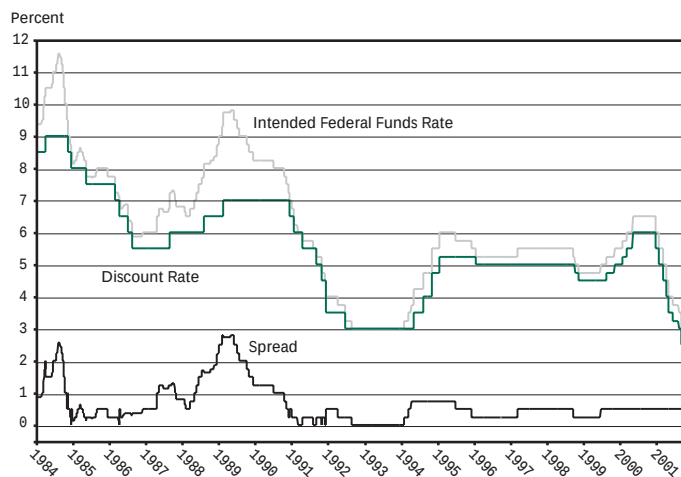


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Conventions used in this publication:

1. Unless otherwise indicated, data are monthly.
2. Shaded areas indicate recessions, as dated by the National Bureau of Economic Research.
3. The *percent change at an annual rate* is the simple, not compounded, monthly percent change multiplied by 12. For example, using consecutive months, the percent change at an annual rate in x between month $t-1$ and the current month t is: $[(x_t / x_{t-1}) - 1] \times 1200$. Note that this differs from *National Economic Trends*. In that publication monthly percent changes are compounded and expressed as annual growth rates.
4. The *percent change from year ago* refers to the percent change from the same period in the previous year. For example, the percent change from year ago in x between month $t-12$ and the current month t is: $[(x_t / x_{t-12}) - 1] \times 100$.

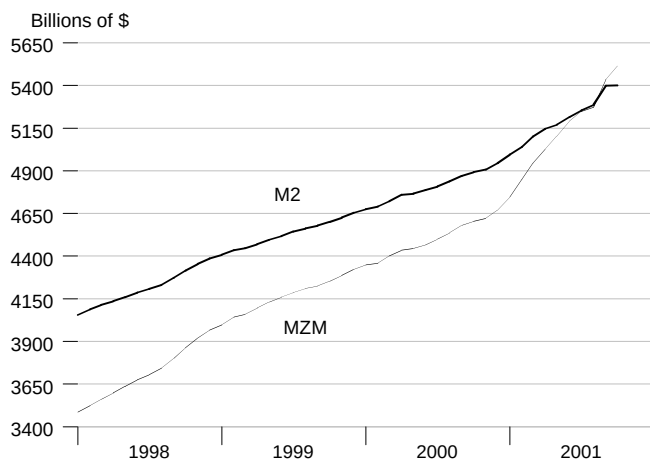
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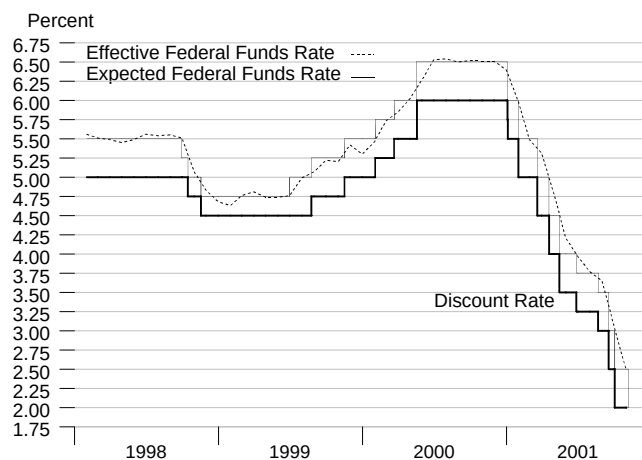
or to:

webmaster@stls.frb.org

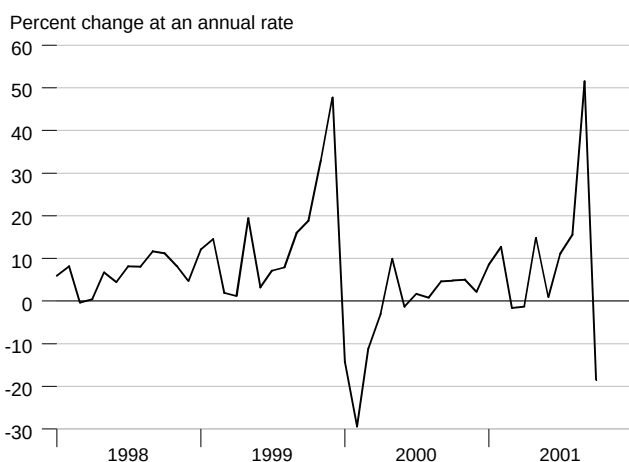
M2 and MZM



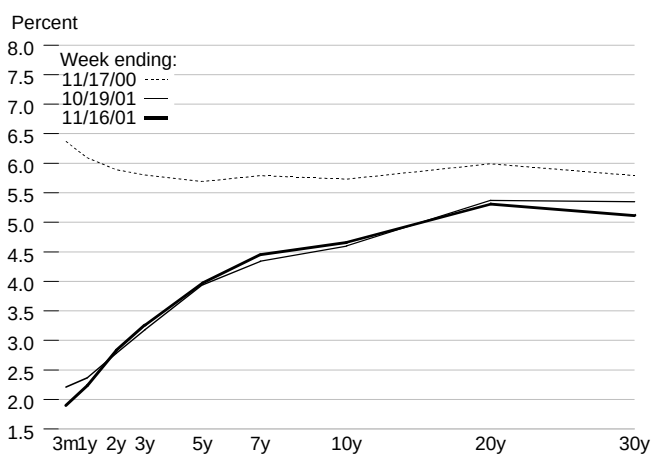
Reserve Market Rates



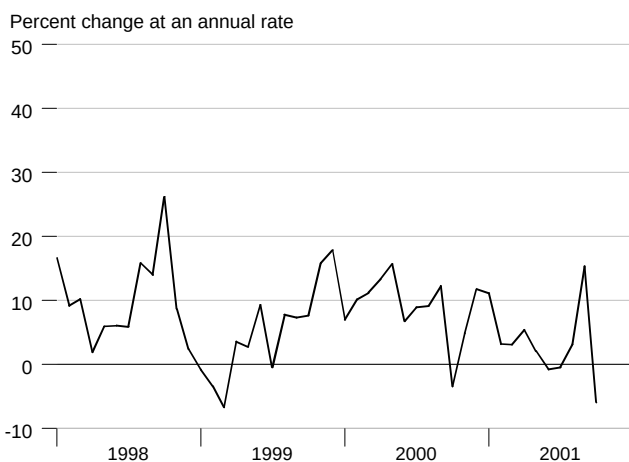
Adjusted Monetary Base



Treasury Yield Curve



Total Bank Credit

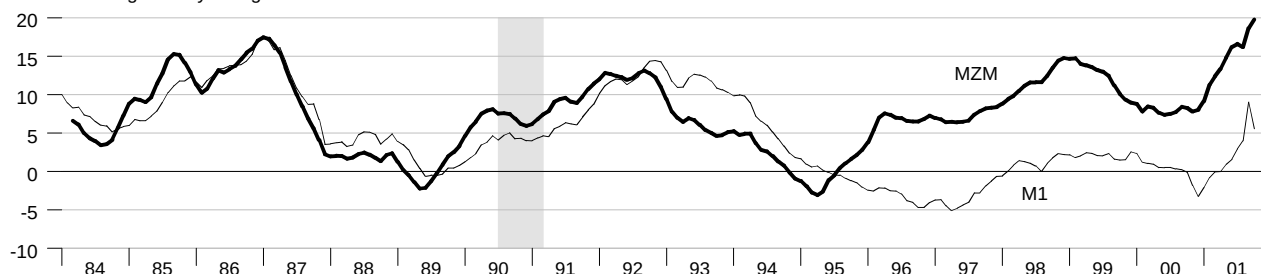


Interest Rates

	Aug 01	Sep 01	Oct 01
Federal Funds Rate	3.65	3.07	2.49
Discount Rate	3.16	2.77	2.02
Prime Rate	6.67	6.28	5.53
Conventional Mortgage Rate	6.95	6.82	6.62
Treasury Yields:			
3-month constant maturity	3.44	2.69	2.20
6-month constant maturity	3.39	2.71	2.17
1-year constant maturity	3.47	2.82	2.33
3-year constant maturity	4.04	3.45	3.14
5-year constant maturity	4.57	4.12	3.91
10-year constant maturity	4.97	4.73	4.57
30-year constant maturity	5.48	5.48	5.32

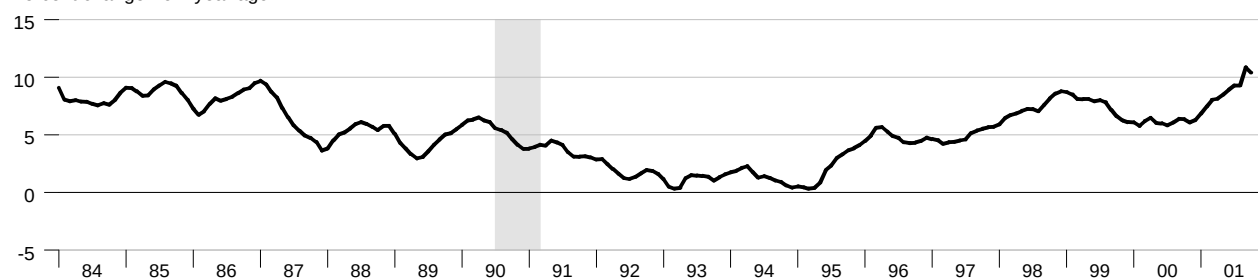
MZM and M1

Percent change from year ago



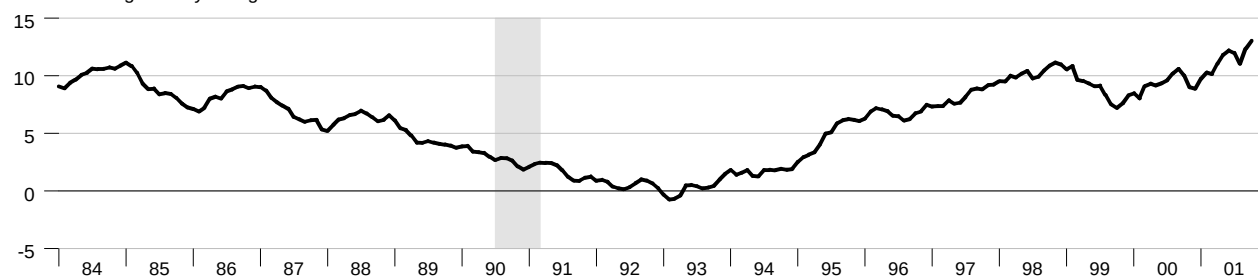
M2

Percent change from year ago



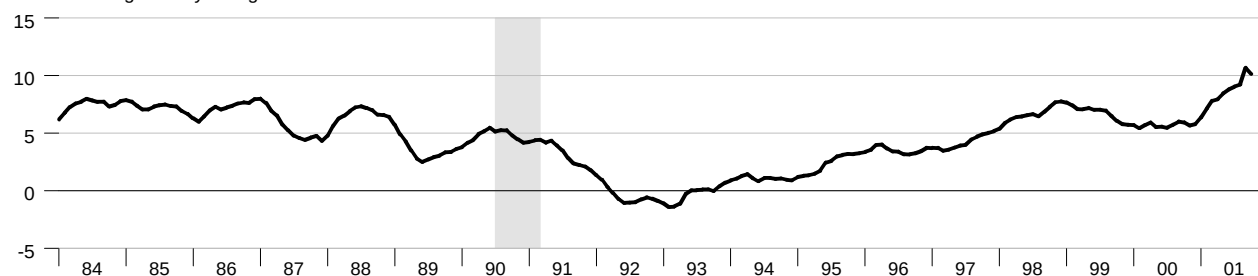
M3

Percent change from year ago



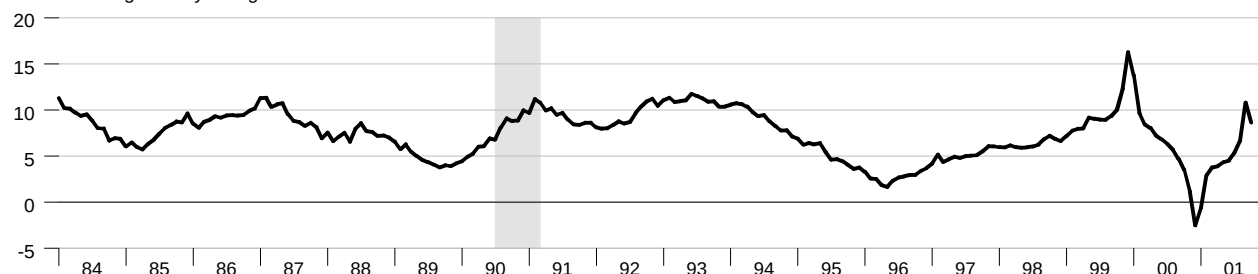
Monetary Services Index - M2

Percent change from year ago



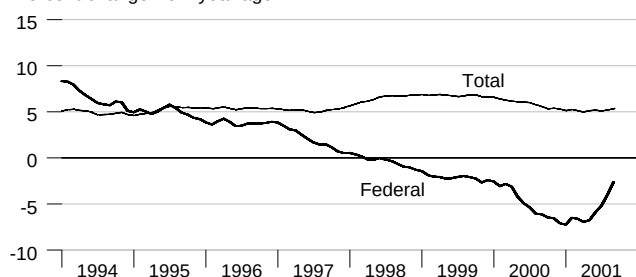
Adjusted Monetary Base

Percent change from year ago



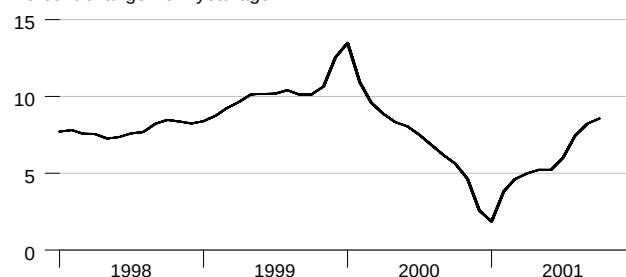
Domestic Nonfinancial Debt

Percent change from year ago



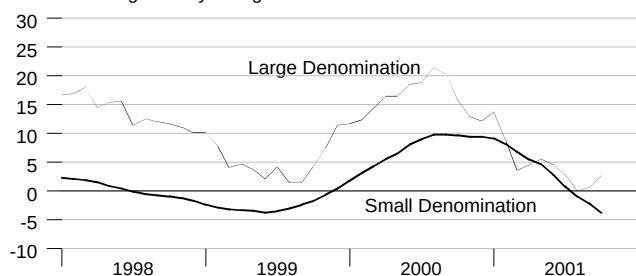
Currency Held by the Nonbank Public

Percent change from year ago



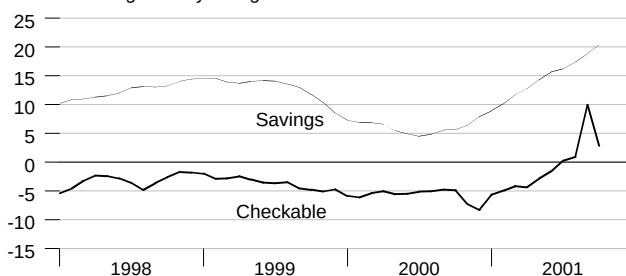
Time Deposits

Percent change from year ago



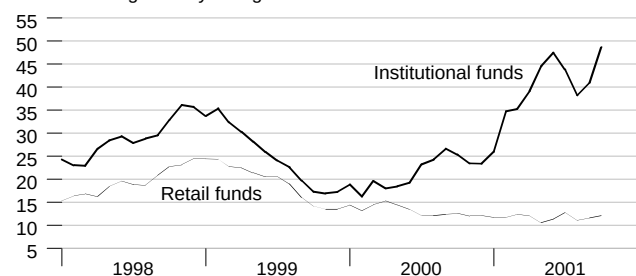
Checkable and Savings Deposits

Percent change from year ago



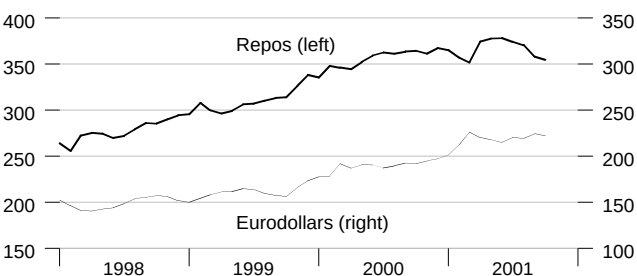
Money Market Mutual Fund Shares

Percent change from year ago



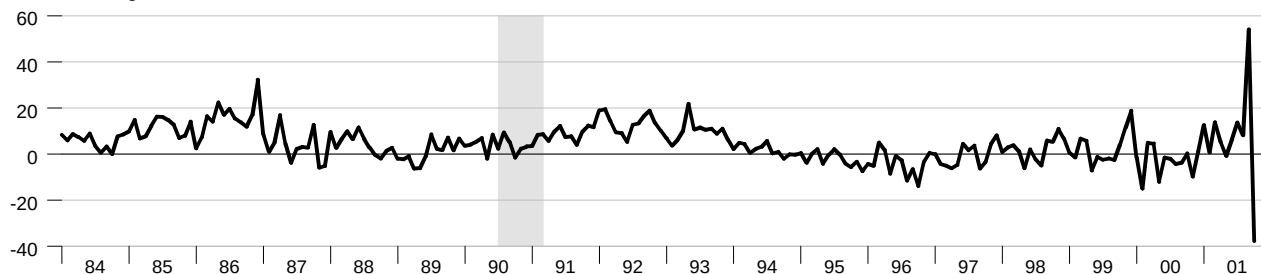
Repurchase Agreements and Eurodollars

Billions of dollars



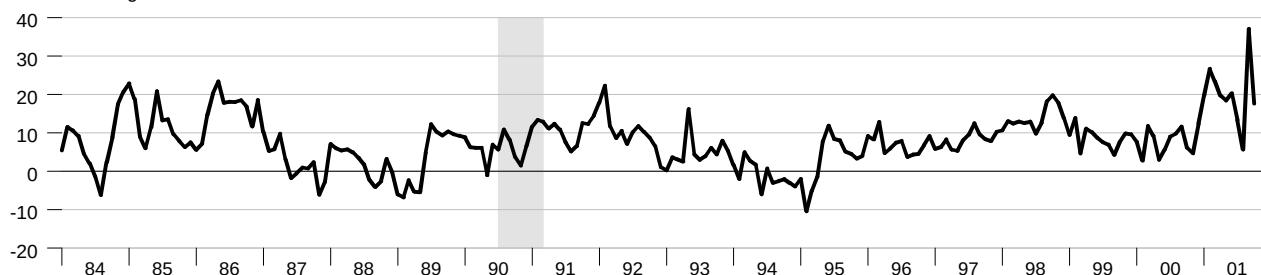
M1

Percent change at an annual rate



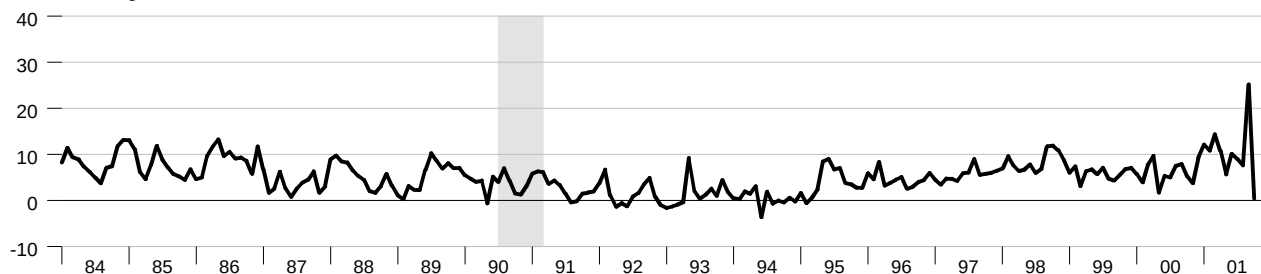
MZM

Percent change at an annual rate



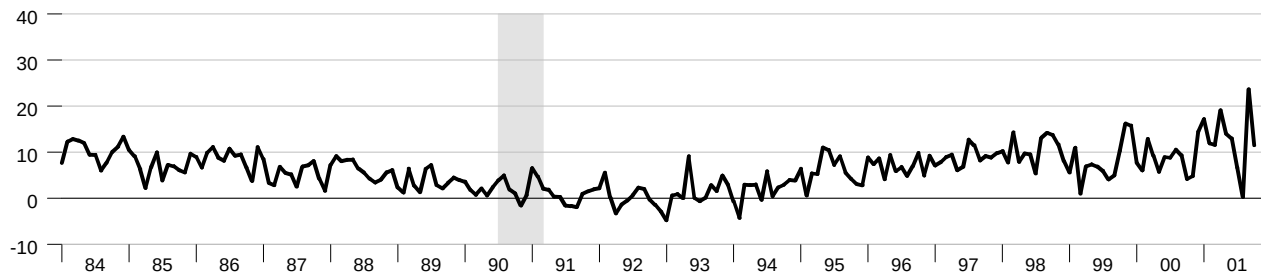
M2

Percent change at an annual rate

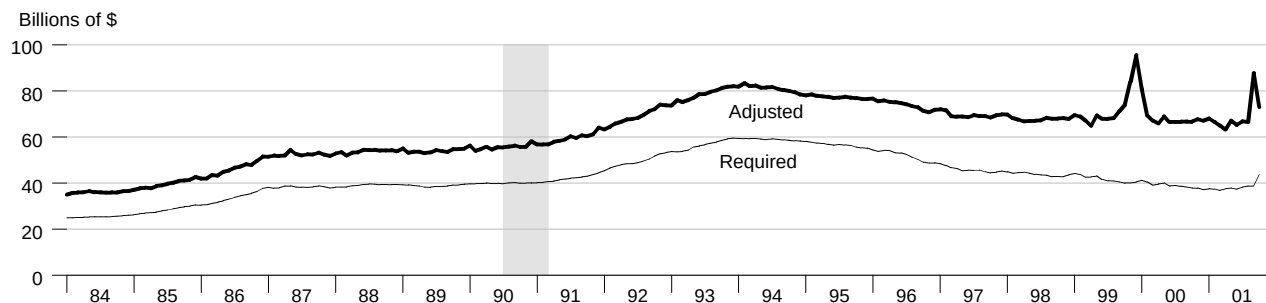


M3

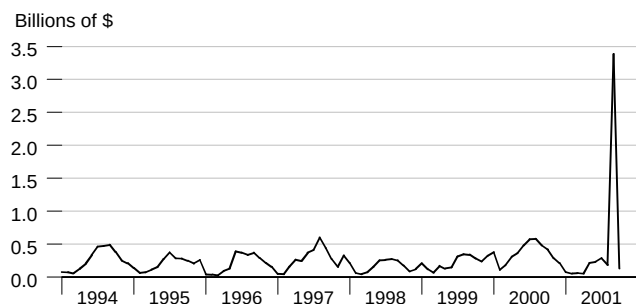
Percent change at an annual rate



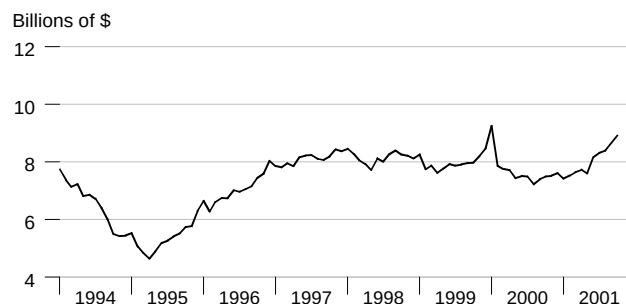
Adjusted and Required Reserves



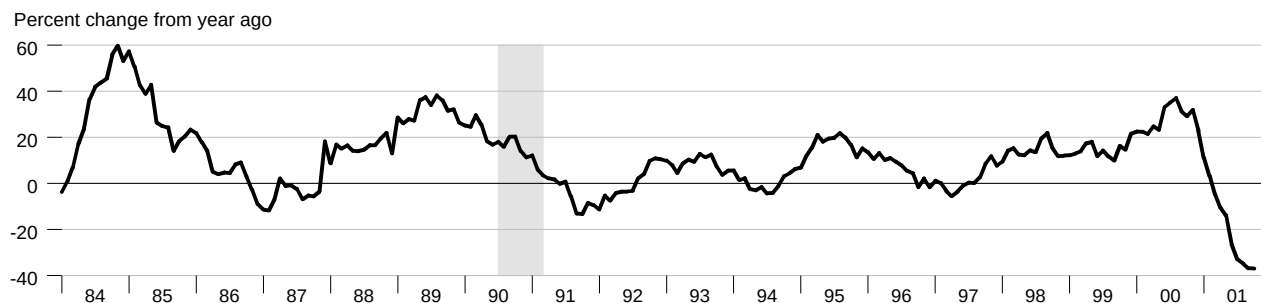
Total Borrowings, nsa



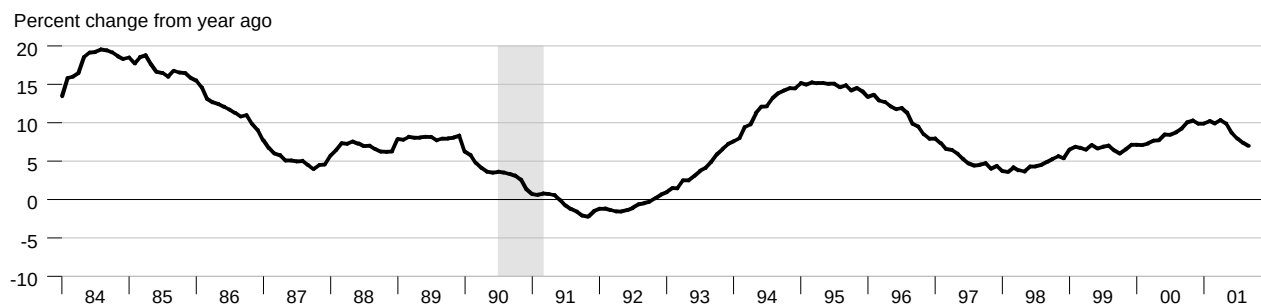
Excess Reserves plus RCB Contracts



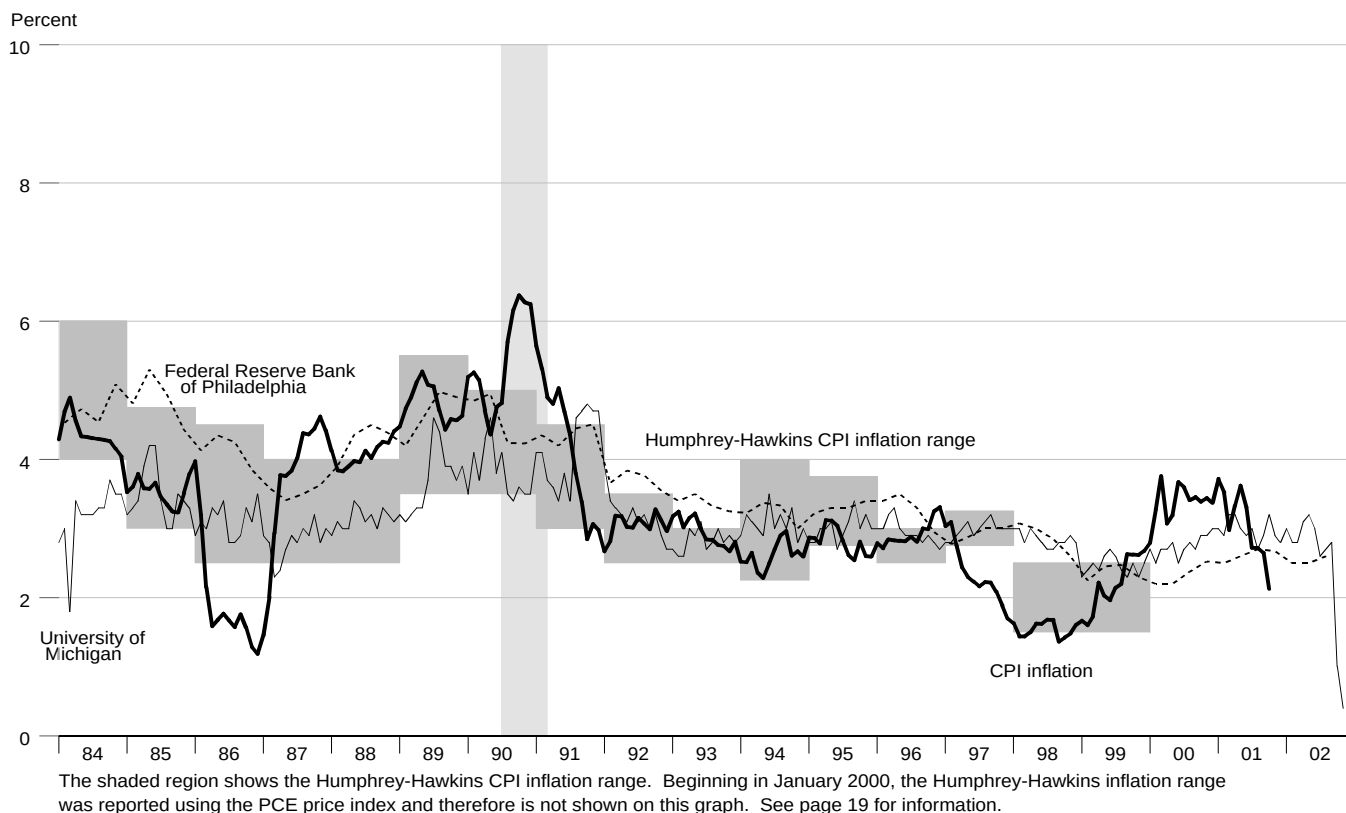
Nonfinancial Commercial Paper



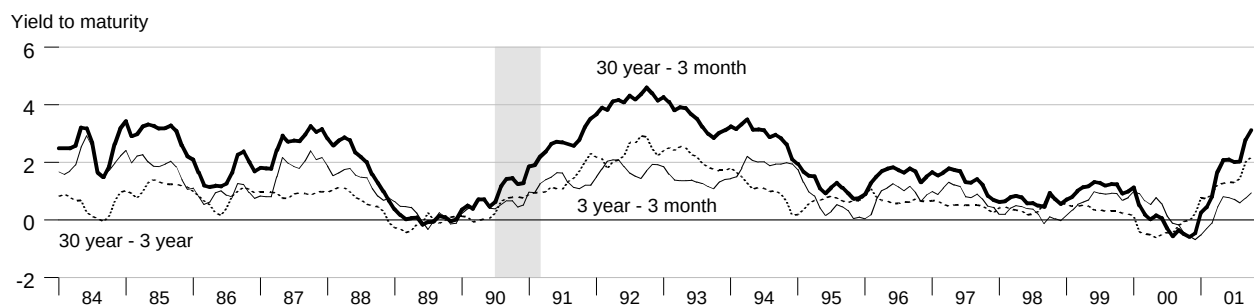
Consumer Credit



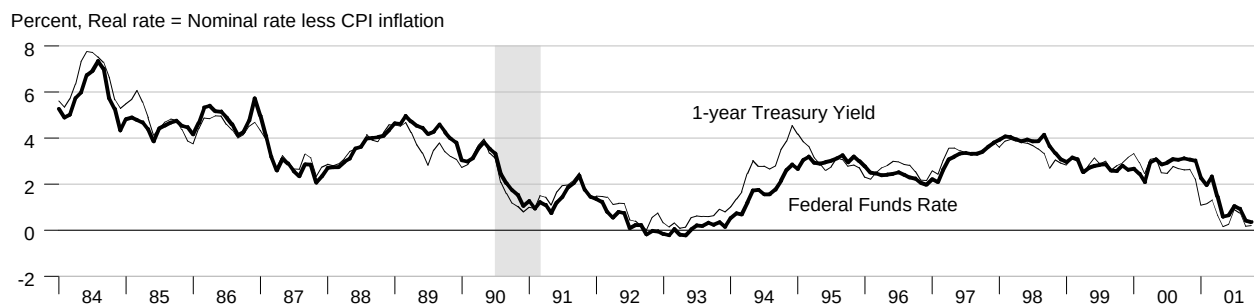
Inflation and Inflation Expectations



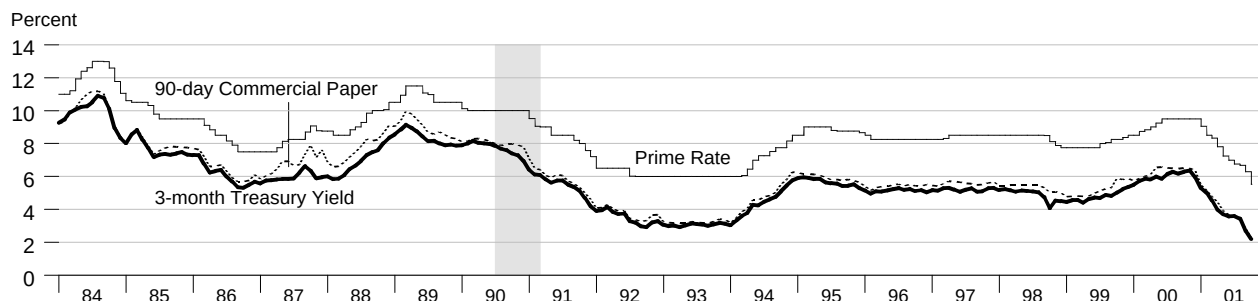
Treasury Security Yield Spreads



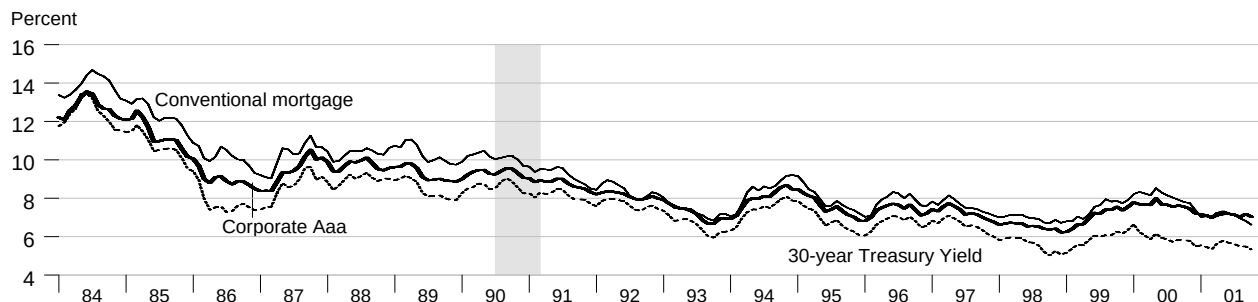
Real Interest Rates



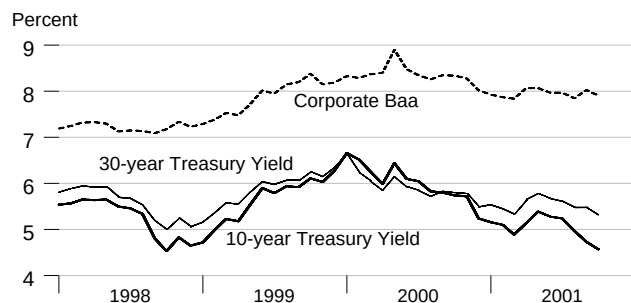
Short Term Interest Rates



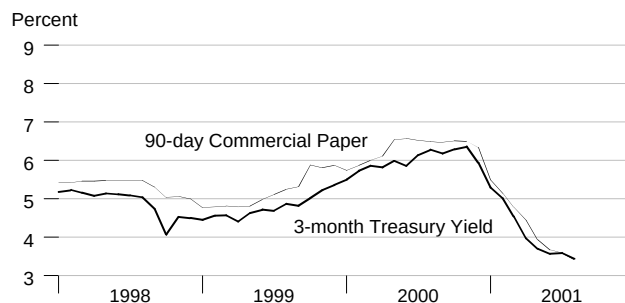
Long Term Interest Rates



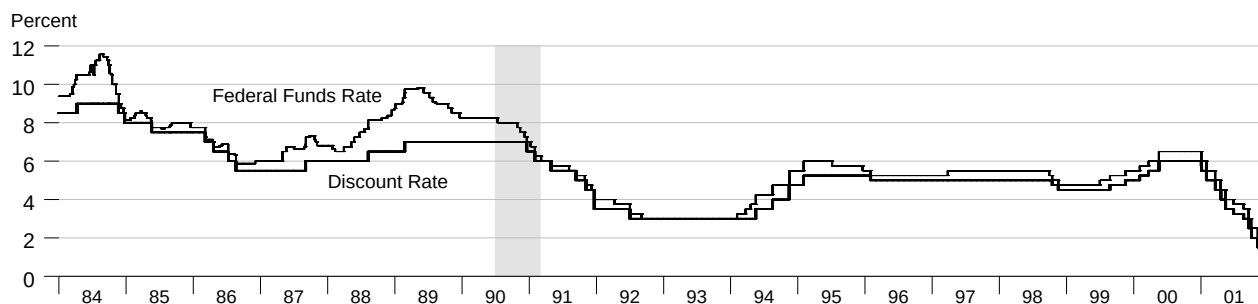
Long Term Interest Rates



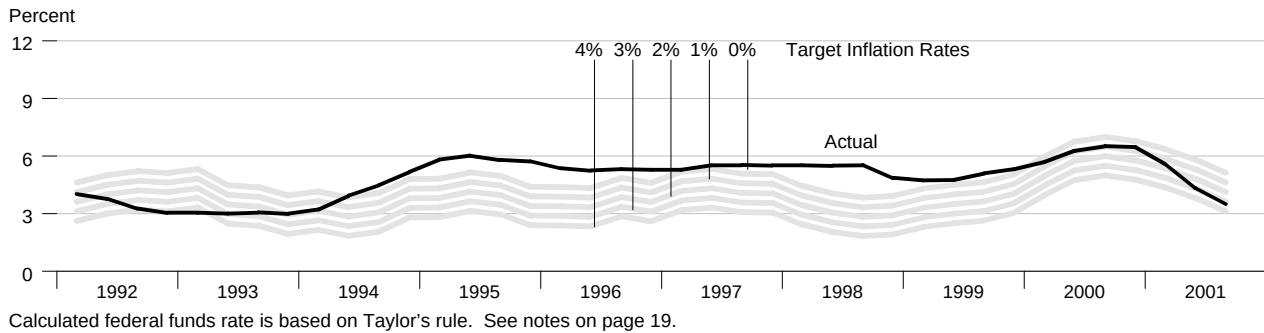
Short Term Interest Rates



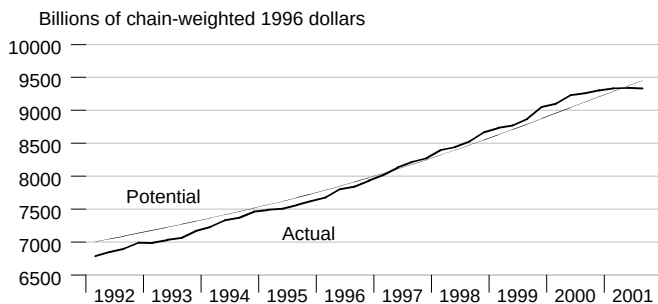
FOMC Expected Federal Funds Rate and Discount Rate



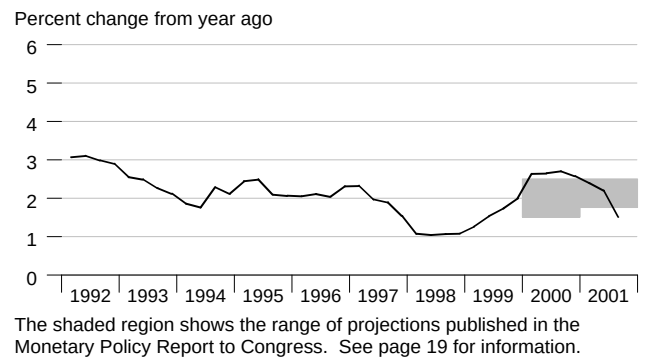
Federal Funds Rate and Inflation Targets



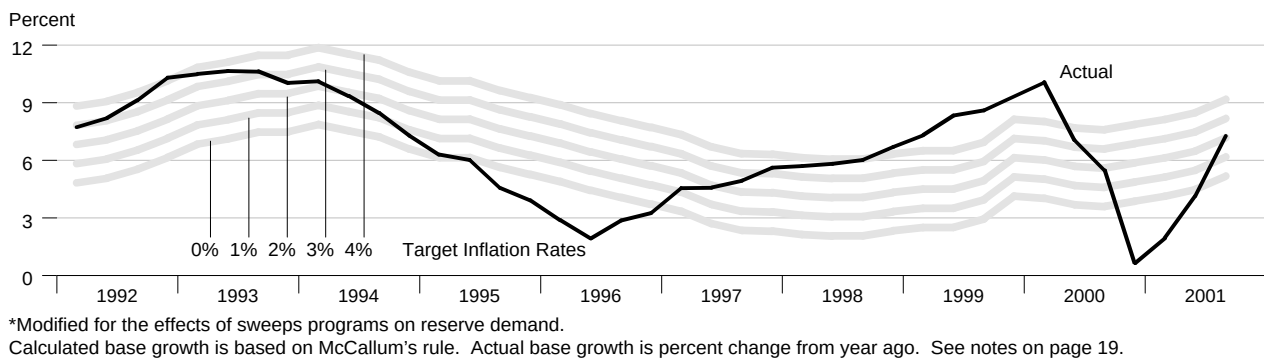
Actual and Potential Real GDP



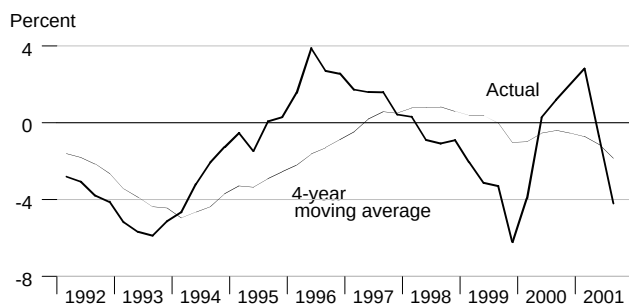
PCE Inflation and Projections



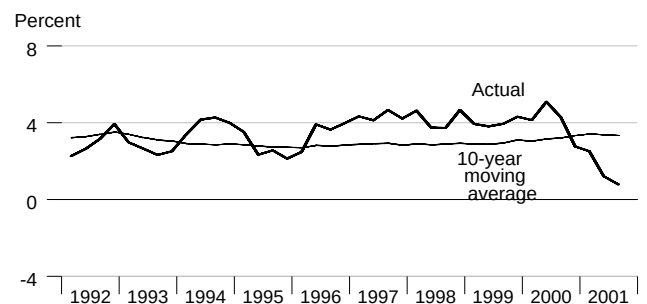
Monetary Base Growth* and Inflation Targets



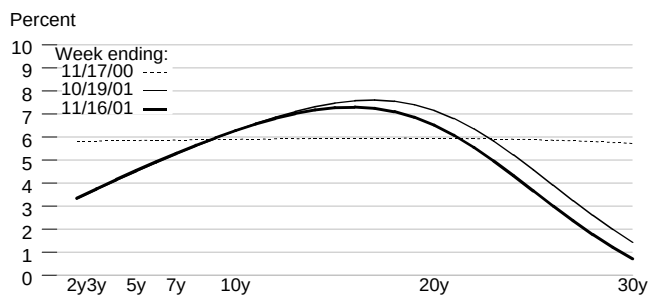
Monetary Base Velocity Growth



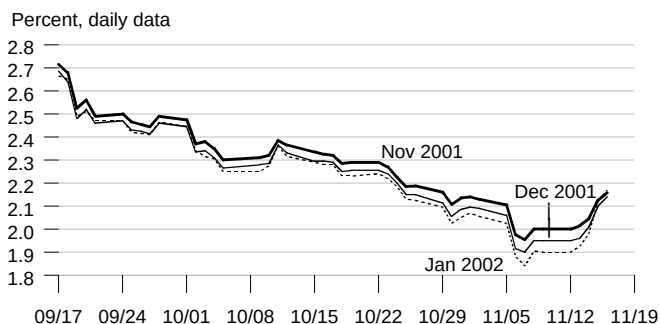
Real Output Growth



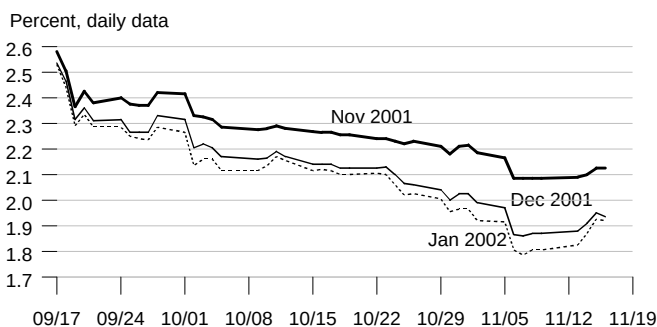
Implied One-Year Forward Rates



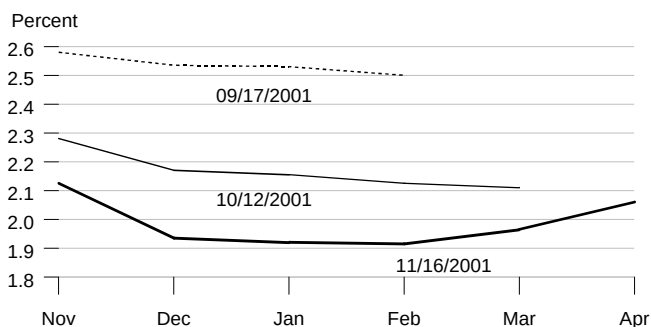
Rates on 3-Month Eurodollar Futures



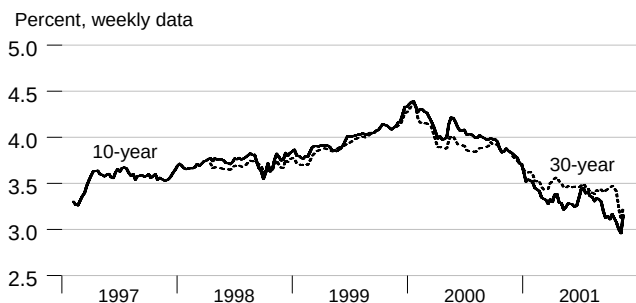
Rates on Selected Fed Funds Futures Contracts



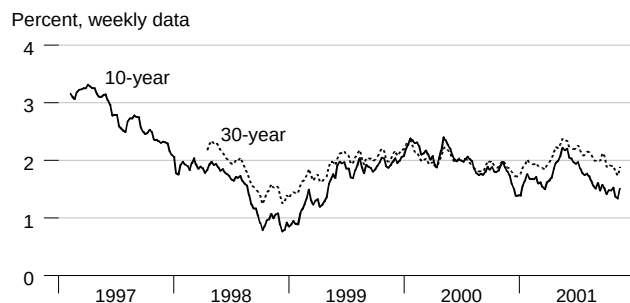
Implied Yields on Fed Funds Futures



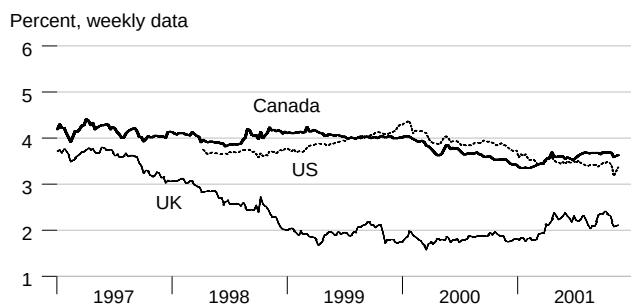
Inflation-Protected Treasury Yields



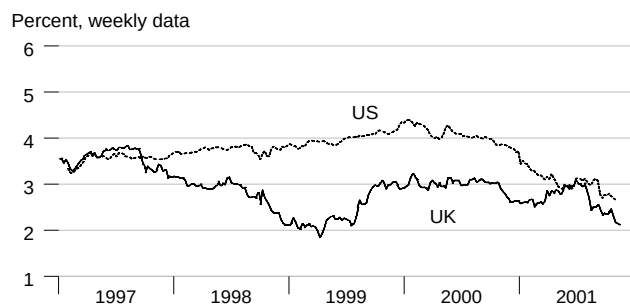
Inflation-Protected Treasury Yield Spreads



Inflation-Indexed 30-Year Bonds



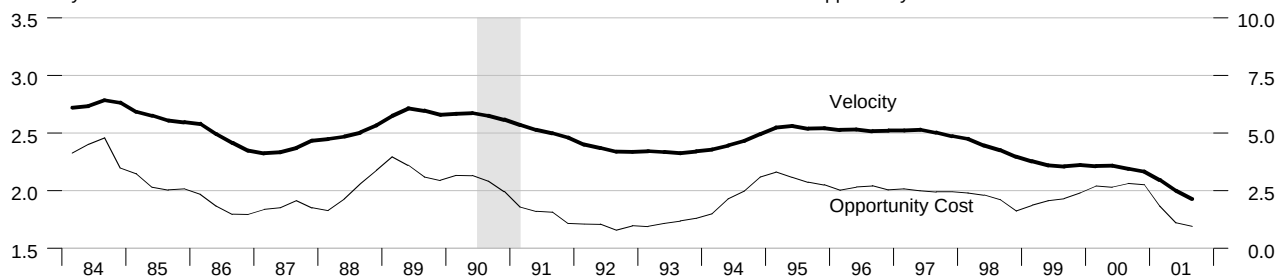
Inflation-Indexed 10-Year Bonds



MZM Velocity and Opportunity Cost

Velocity = Nominal GDP / MZM

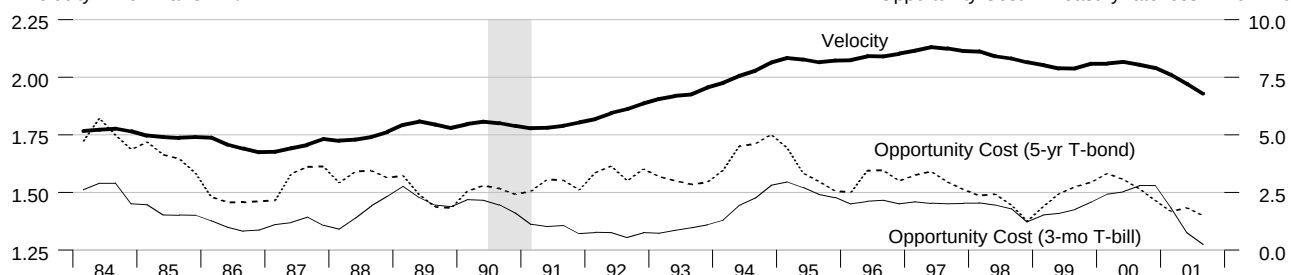
Opportunity Cost = 3 month T-bill rate less MZM own rate



M2 Velocity and Opportunity Cost

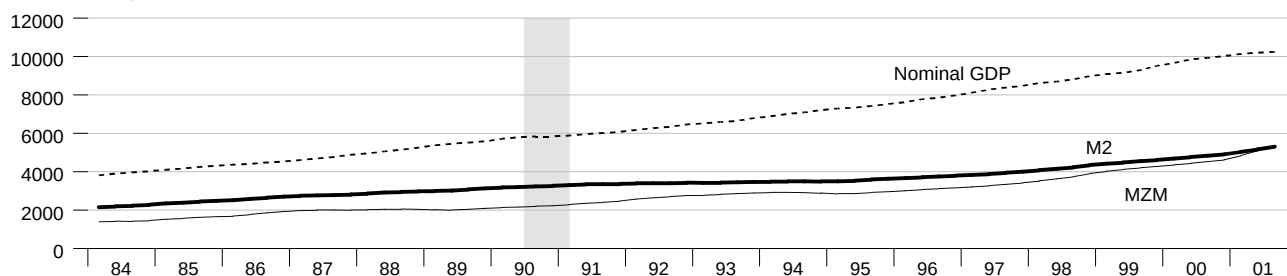
Velocity = Nominal GDP / M2

Opportunity Cost = Treasury rate less M2 own rate



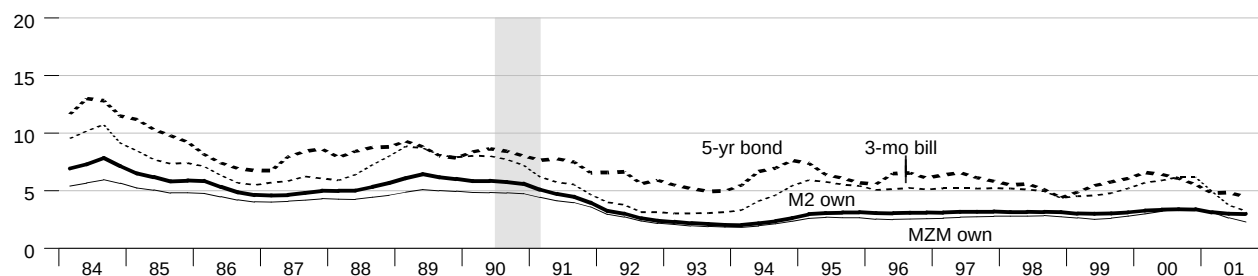
M2, MZM and Nominal GDP

Billions of \$



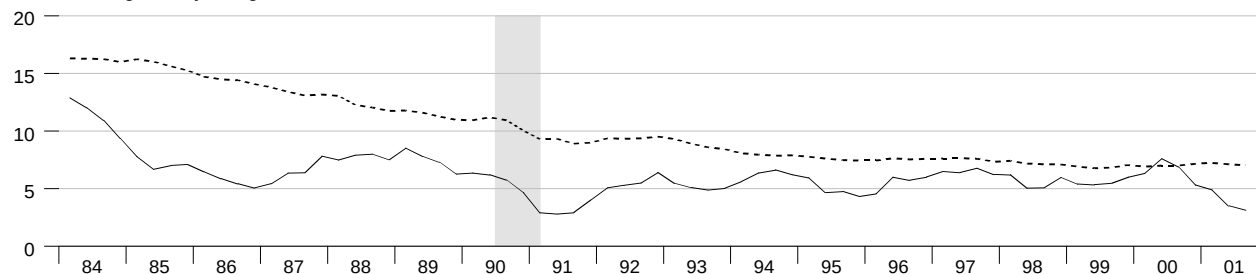
Interest Rates

Percent



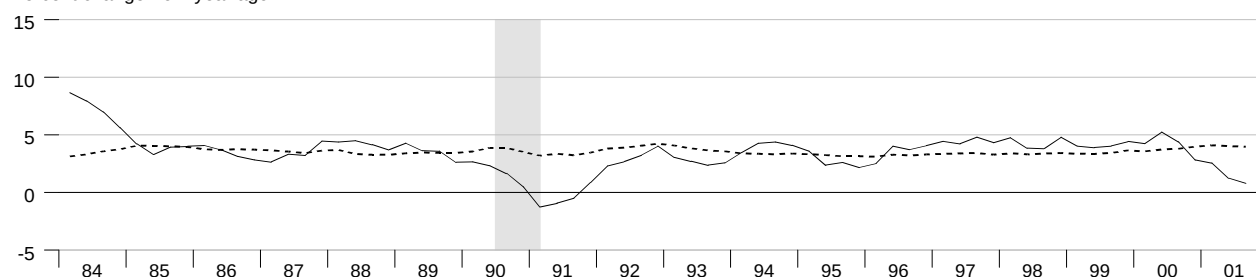
Gross Domestic Product

Percent change from year ago



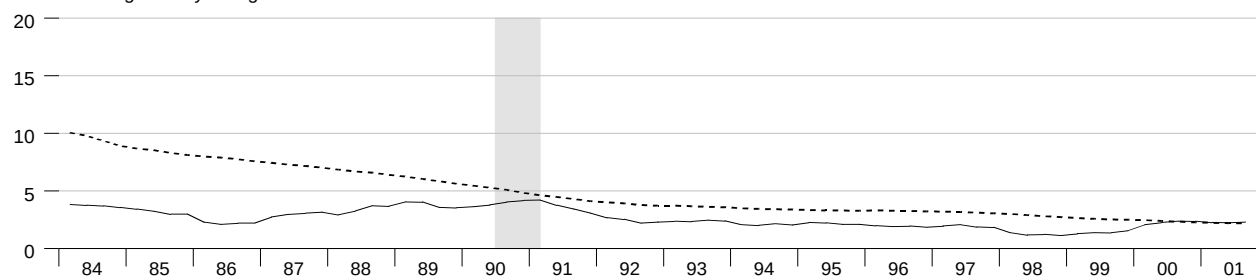
Real Gross Domestic Product

Percent change from year ago



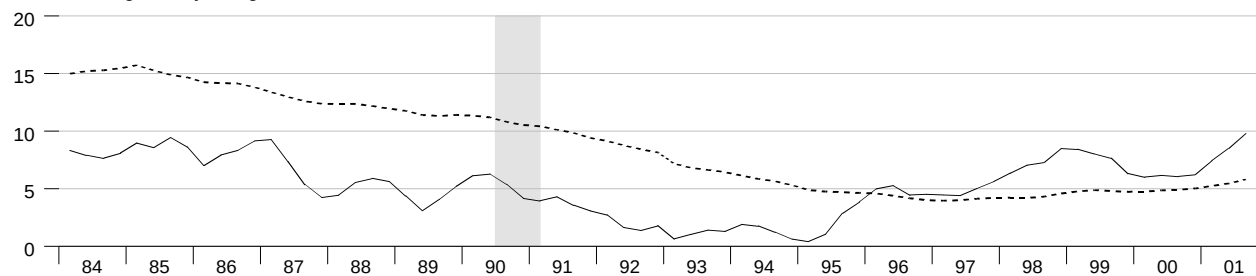
Gross Domestic Product Price Index

Percent change from year ago



M2

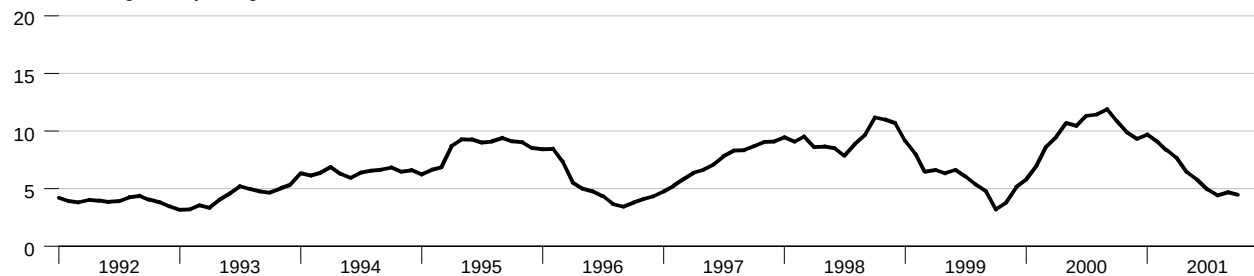
Percent change from year ago



Dashed lines indicate 10-year moving averages

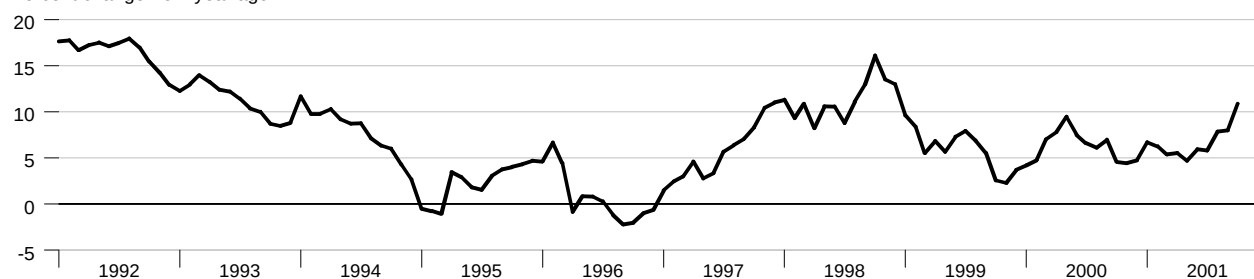
Bank Credit

Percent change from year ago



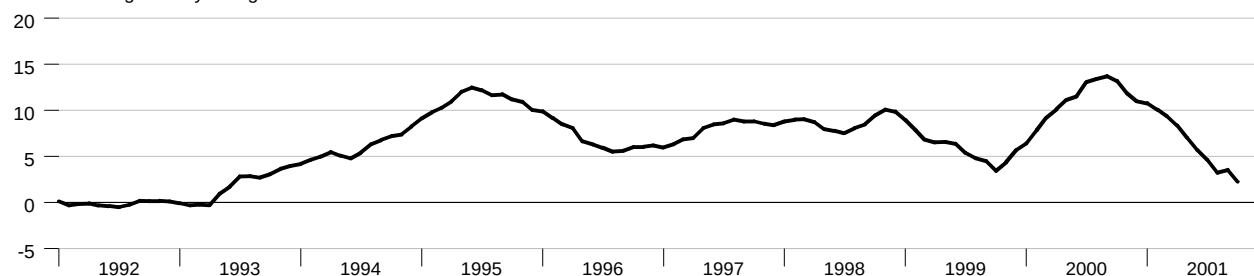
Investment Securities in Bank Credit at Commercial Banks

Percent change from year ago



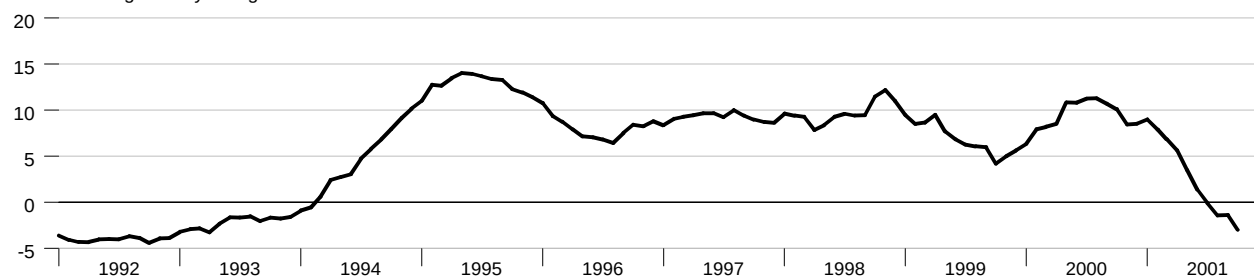
Total Loans and Leases in Bank Credit at Commercial Banks

Percent change from year ago

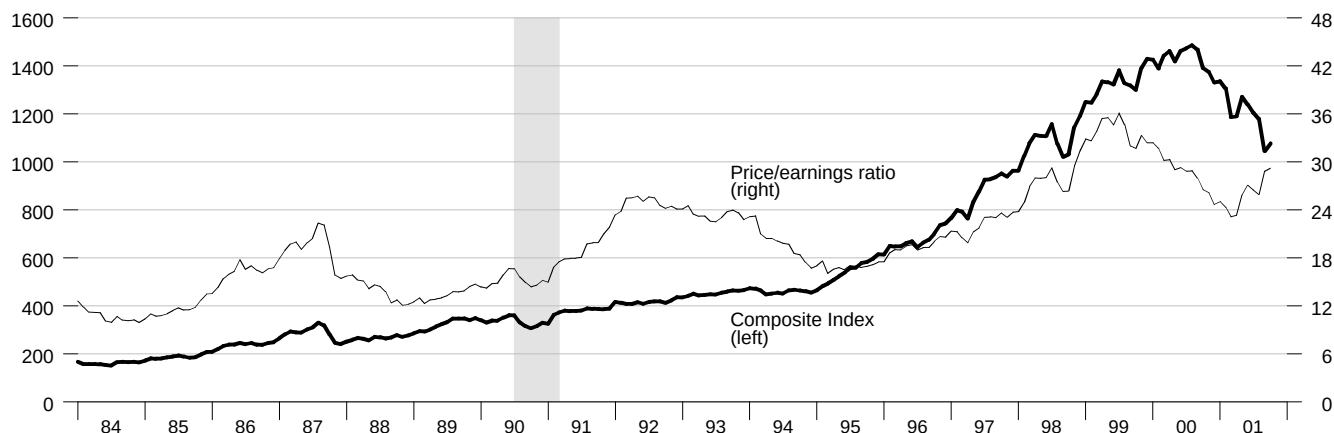


Commercial and Industrial Loans at Commercial Banks

Percent change from year ago



Standard and Poor's 500



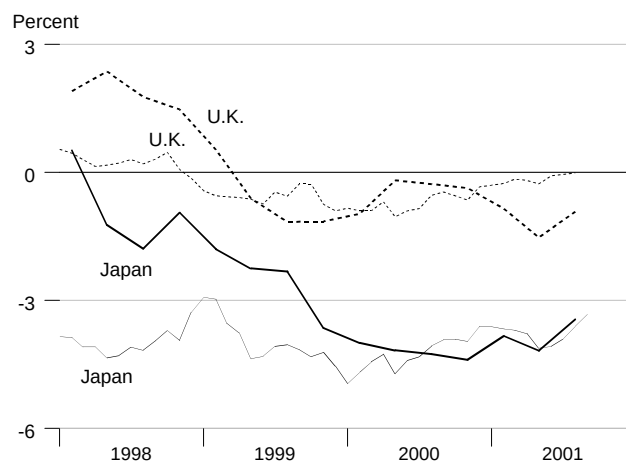
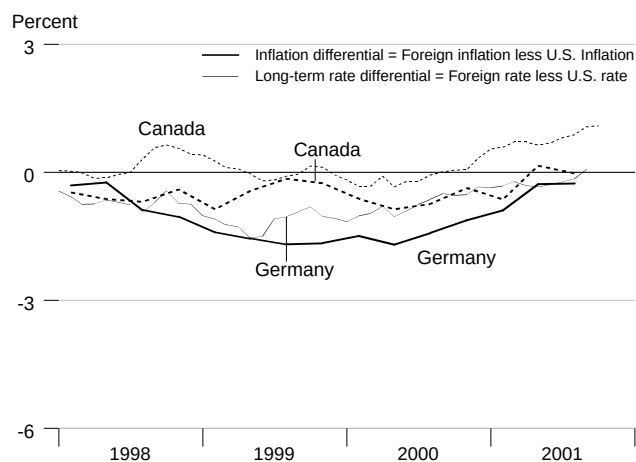
Inflation and Long-Term Interest Rates

Trend in Consumer Price Inflation Rates
Percent change from year ago

Recent Long-Term Government Bond Rates
Percent

	2000Q4	2001Q1	2001Q2	2001Q3	Jul01	Aug01	Sep01	Oct01
United States	3.44	3.41	3.44	2.72	5.24	4.97	4.73	4.57
Canada	3.08	2.77	3.60	2.69	6.05	5.85	5.80	5.66
France	1.89	1.29	2.02	1.79	5.46	5.29	5.26	.
Germany	2.32	2.52	3.16	2.46	5.02	4.82	4.81	.
Italy	2.67	2.89	3.05	2.80	5.40	5.22	5.20	.
Japan	-0.95	-0.43	-0.74	-0.73	1.33	1.36	1.40	.
United Kingdom	3.07	2.55	1.91	1.80	5.19	4.96	.	.

Inflation and Long-Term Interest Rates Differentials



		Money Stock				Bank			
		M1	MZM	M2	M3	Credit	Monetary Base	Reserves	MSI M2
1996		1105.818	3096.125	3739.297	4811.846	3685.255	455.572	73.952	217.463
1997		1069.145	3318.867	3921.981	5206.953	3953.511	478.708	69.523	226.608
1998		1079.795	3706.274	4208.613	5740.107	4326.598	508.942	67.808	241.647
1999		1101.661	4163.696	4527.993	6251.212	4582.464	557.865	72.360	258.034
2000		1103.866	4496.753	4805.105	6834.241	5028.924	590.821	68.319	272.755
1999	1	1098.625	4032.495	4429.975	6094.477	4517.056	536.334	68.521	252.787
	2	1102.740	4128.668	4494.332	6188.695	4527.725	545.912	67.392	256.223
	3	1095.559	4207.215	4561.360	6278.751	4581.633	557.969	69.050	259.750
	4	1109.718	4286.407	4626.303	6442.925	4703.440	591.246	84.477	263.377
2000	1	1115.416	4368.912	4695.981	6615.463	4840.144	593.102	72.390	266.963
	2	1109.967	4448.594	4771.350	6763.296	4990.219	586.045	67.097	270.750
	3	1099.561	4538.641	4838.632	6915.539	5110.887	589.054	66.636	274.657
	4	1090.519	4630.863	4914.457	7042.665	5174.444	595.084	67.150	278.650
2001	1	1104.185	4846.698	5045.395	7281.324	5277.215	604.850	66.513	285.920
	2	1119.064	5105.942	5177.080	7554.335	5319.554	610.943	65.171	293.497
	3	1157.730	5317.129	5312.917	7732.002	5350.055	633.751	73.636	301.147
1999	Oct	1096.970	4251.923	4599.722	6357.521	4639.456	572.990	73.928	261.920
	Nov	1107.434	4286.589	4625.905	6443.344	4700.546	588.675	84.023	263.320
	Dec	1124.750	4320.709	4653.281	6527.911	4770.318	612.073	95.479	264.890
2000	Jan	1123.267	4348.064	4675.512	6569.730	4798.061	604.796	80.824	266.040
	Feb	1109.242	4358.041	4690.942	6602.927	4838.623	589.984	69.258	266.710
	Mar	1113.739	4400.631	4721.490	6673.732	4883.749	584.525	67.089	268.140
	Apr	1117.934	4434.341	4759.663	6725.071	4937.886	583.053	65.913	270.090
	May	1106.712	4445.362	4766.589	6757.266	5002.355	587.863	68.889	270.450
	Jun	1105.255	4466.080	4787.799	6807.550	5030.415	587.220	66.490	271.710
	Jul	1103.351	4499.522	4807.879	6857.486	5067.805	588.032	66.555	272.930
	Aug	1099.380	4536.293	4838.017	6917.705	5106.393	588.435	66.664	274.640
	Sep	1095.953	4580.107	4870.001	6971.426	5158.463	590.694	66.689	276.400
	Oct	1096.147	4603.242	4891.430	6995.722	5143.717	593.064	66.687	277.470
	Nov	1087.217	4621.121	4906.816	7023.776	5164.541	595.549	67.685	278.240
	Dec	1088.194	4668.227	4945.124	7108.498	5215.074	596.639	67.079	280.240
2001	Jan	1099.502	4744.973	4995.163	7210.177	5263.401	600.887	67.999	282.980
	Feb	1100.229	4850.193	5040.334	7281.767	5277.319	607.236	66.558	285.720
	Mar	1112.823	4944.929	5100.687	7352.027	5290.925	606.426	64.981	289.060
	Apr	1117.625	5025.857	5146.294	7468.912	5314.652	605.803	63.241	291.580
	May	1116.883	5102.939	5170.661	7556.165	5323.747	613.264	67.022	293.300
	Jun	1122.685	5189.030	5214.285	7637.927	5320.264	613.761	65.249	295.610
	Jul	1135.426	5246.470	5253.526	7680.257	5318.151	619.434	66.748	297.600
	Aug	1143.153	5271.160	5287.106	7682.236	5331.985	627.448	66.472	299.950
	Sep	1194.612	5433.757	5398.118	7833.513	5400.029	654.372	87.688	305.890
	Oct	1157.017	5513.381	5400.086	7908.605	5373.311	644.274	72.979	305.610

*All values are given in billions of dollars

		Federal Funds	Discount Rate	Prime Rate	3-mo CDs	Treasury Yields			Corporate Aaa Bonds	S & L Aaa Bonds	Conventional Mortgage
						3 mo	3 yr	30 yr			
1996		5.30	5.02	8.27	5.39	5.15	5.99	6.70	7.37	5.52	7.80
	1997	5.46	5.00	8.44	5.62	5.20	6.10	6.61	7.26	5.32	7.60
	1998	5.35	4.92	8.35	5.47	4.91	5.14	5.58	6.53	4.93	6.94
	1999	4.97	4.62	7.99	5.33	4.78	5.49	5.87	7.04	5.28	7.43
	2000	6.24	5.73	9.23	6.46	6.00	6.22	5.94	7.62	5.58	8.06
1999	1	4.73	4.50	7.75	4.90	4.53	4.87	5.37	6.42	4.87	6.88
	2	4.75	4.50	7.75	4.98	4.59	5.35	5.80	6.93	5.05	7.20
	3	5.09	4.60	8.10	5.38	4.79	5.71	6.04	7.33	5.42	7.80
	4	5.31	4.87	8.37	6.06	5.20	6.00	6.25	7.49	5.79	7.83
	2000	5.68	5.19	8.69	6.03	5.70	6.56	6.30	7.71	5.82	8.26
	2	6.27	5.74	9.25	6.57	5.89	6.52	5.98	7.77	5.72	8.32
	3	6.52	6.00	9.50	6.63	6.20	6.16	5.80	7.61	5.45	8.03
	4	6.47	6.00	9.50	6.59	6.20	5.63	5.69	7.40	5.32	7.64
	2001	5.59	5.11	8.62	5.26	4.95	4.64	5.44	7.08	5.03	7.01
	2	4.33	3.83	7.34	4.10	3.75	4.43	5.70	7.22	5.11	7.13
	3	3.50	3.06	6.57	3.34	3.24	3.93	5.52	7.11	4.87	6.97
1999	Oct	5.20	4.75	8.25	6.13	5.02	5.94	6.26	7.55	5.78	7.85
	Nov	5.42	4.86	8.37	6.00	5.23	5.92	6.15	7.36	5.77	7.74
	Dec	5.30	5.00	8.50	6.05	5.36	6.14	6.35	7.55	5.82	7.91
2000	Jan	5.46	5.00	8.50	5.95	5.50	6.49	6.63	7.78	5.91	8.21
	Feb	5.73	5.24	8.73	6.01	5.73	6.65	6.23	7.68	5.88	8.33
	Mar	5.85	5.34	8.83	6.14	5.86	6.53	6.05	7.68	5.68	8.24
	Apr	6.02	5.50	9.00	6.28	5.82	6.36	5.85	7.64	5.60	8.15
	May	6.27	5.71	9.24	6.71	5.99	6.77	6.15	7.99	5.87	8.52
	Jun	6.53	6.00	9.50	6.73	5.86	6.43	5.93	7.67	5.69	8.29
	Jul	6.54	6.00	9.50	6.67	6.14	6.28	5.85	7.65	5.53	8.15
	Aug	6.50	6.00	9.50	6.61	6.28	6.17	5.72	7.55	5.43	8.03
	Sep	6.52	6.00	9.50	6.60	6.18	6.02	5.83	7.62	5.40	7.91
	Oct	6.51	6.00	9.50	6.67	6.29	5.85	5.80	7.55	5.46	7.80
	Nov	6.51	6.00	9.50	6.65	6.36	5.79	5.78	7.45	5.38	7.75
	Dec	6.40	6.00	9.50	6.45	5.94	5.26	5.49	7.21	5.11	7.38
2001	Jan	5.98	5.52	9.05	5.62	5.29	4.77	5.54	7.15	4.99	7.03
	Feb	5.49	5.00	8.50	5.26	5.01	4.71	5.45	7.10	5.09	7.05
	Mar	5.31	4.81	8.32	4.89	4.54	4.43	5.34	6.98	5.00	6.95
	Apr	4.80	4.28	7.80	4.53	3.97	4.42	5.65	7.20	5.14	7.08
	May	4.21	3.73	7.24	4.02	3.70	4.51	5.78	7.29	5.15	7.15
	Jun	3.97	3.47	6.98	3.74	3.57	4.35	5.67	7.18	5.03	7.16
	Jul	3.77	3.25	6.75	3.66	3.59	4.31	5.61	7.13	4.79	7.13
	Aug	3.65	3.16	6.67	3.48	3.44	4.04	5.48	7.02	4.89	6.95
	Sep	3.07	2.77	6.28	2.87	2.69	3.45	5.48	7.17	4.93	6.82
	Oct	2.49	2.02	5.53	2.31	2.20	3.14	5.32	7.03		6.62

*All values are given as a percent at an annual rate

		M1	MZM	M2	M3
Percent change from previous period					
1996		-3.21	6.56	4.80	6.75
1997		-3.32	7.19	4.89	8.21
1998		1.00	11.67	7.31	10.24
1999		2.02	12.34	7.59	8.90
2000		0.20	8.00	6.12	9.33
1999					
	1	0.83	2.97	1.80	1.89
	2	0.37	2.38	1.45	1.55
	3	-0.65	1.90	1.49	1.46
	4	1.29	1.88	1.42	2.61
2000					
	1	0.51	1.92	1.51	2.68
	2	-0.49	1.82	1.60	2.23
	3	-0.94	2.02	1.41	2.25
	4	-0.82	2.03	1.57	1.84
2001					
	1	1.25	4.66	2.66	3.39
	2	1.35	5.35	2.61	3.75
	3	3.46	4.14	2.62	2.35
1999					
	Oct	0.33	0.63	0.46	0.86
	Nov	0.95	0.82	0.57	1.35
	Dec	1.56	0.80	0.59	1.31
2000					
	Jan	-0.13	0.63	0.48	0.64
	Feb	-1.25	0.23	0.33	0.51
	Mar	0.41	0.98	0.65	1.07
	Apr	0.38	0.77	0.81	0.77
	May	-1.00	0.25	0.15	0.48
	Jun	-0.13	0.47	0.44	0.74
	Jul	-0.17	0.75	0.42	0.73
	Aug	-0.36	0.82	0.63	0.88
	Sep	-0.31	0.97	0.66	0.78
	Oct	0.02	0.51	0.44	0.35
	Nov	-0.81	0.39	0.31	0.40
	Dec	0.09	1.02	0.78	1.21
2001					
	Jan	1.04	1.64	1.01	1.43
	Feb	0.07	2.22	0.90	0.99
	Mar	1.14	1.95	1.20	0.96
	Apr	0.43	1.64	0.89	1.59
	May	-0.07	1.53	0.47	1.17
	Jun	0.52	1.69	0.84	1.08
	Jul	1.13	1.11	0.75	0.55
	Aug	0.68	0.47	0.64	0.03
	Sep	4.50	3.08	2.10	1.97
	Oct	-3.15	1.47	0.04	0.96

Definitions

M1: The sum of currency held outside the vaults of depository institutions, Federal Reserve Banks, and the U.S. Treasury; travelers checks; and demand and other checkable deposits issued by financial institutions (except demand deposits due to the Treasury and depository institutions), minus cash items in process of collection and Federal Reserve float.

MZM: M2 minus small denomination time deposits, plus institutional money market mutual funds. The label MZM was coined by William Poole (1991) for this aggregate, proposed earlier by Motley (1988).

M2: M1 plus savings deposits (including money market deposit accounts) and small-denomination (less than \$100,000) time deposits issued by financial institutions; and shares in retail money market mutual funds (funds with initial investments of less than \$50,000), net of retirement accounts.

M3: M2 plus large-denomination (\$100,000 or more) time deposits; repurchase agreements issued by depository institutions; Eurodollar deposits, specifically, dollar-denominated deposits due to nonbank U.S. addresses held at foreign offices of U.S. banks worldwide and all banking offices in Canada and the United Kingdom; and institutional money market mutual funds (funds with initial investments of \$50,000 or more).

Bank Credit: All loans, leases, and securities held by commercial banks.

Domestic Nonfinancial Debt: Total credit market liabilities of the U.S. Treasury, federally sponsored agencies, state and local governments, households, and firms (except depository institutions and money market mutual funds).

Adjusted Monetary Base: The sum of currency in circulation outside Federal Reserve Banks and the U.S. Treasury, deposits of depository financial institutions at Federal Reserve Banks, and an adjustment for the effects of changes in statutory reserve requirements on the quantity of base money held by depositories. This series is a spliced chain index; see Anderson and Rasche (1996a,b).

Adjusted Reserves: The sum of vault cash and Federal Reserve Bank deposits held by depository institutions and an adjustment for the effects of changes in statutory reserve requirements on the quantity of base money held by depositories. This series, a spliced chain index, is numerically larger than the Board of Governors' measure, which excludes vault cash not used to satisfy statutory reserve requirements and Federal Reserve Bank deposits used to satisfy required clearing balance contracts; see Anderson and Rasche (1996a) and www.stls.frb.org/research/newbase.html.

Monetary Services Index: An index that measures the flow of monetary services received by households and firms from their holdings of liquid assets; see Anderson, Jones, and Nesmith (1997). Indexes are shown for the assets included in M2; additional data are available at www.stls.frb.org/research/msi/index.html.

Note: M1, M2, M3, Bank Credit, and Domestic Nonfinancial Debt are constructed and published by the Board of Governors of the Federal Reserve System. For details, see *Federal Reserve Bulletin*, tables 1.21 and 1.26. MZM, Adjusted Monetary Base, Adjusted Reserves, and Monetary Services Index are constructed and published by the Research Division of the Federal Reserve Bank of St. Louis.

Notes

Page 3: **MZM**, or "Money, Zero Maturity," includes the zero maturity, or immediately available, components of M3. MZM equals M2 minus small-denomination time deposits, plus institutional money market mutual funds (that is, the money market mutual funds included in M3 but excluded from M2). Readers are cautioned that since early 1994 the level and growth of M1 have been depressed by retail sweep programs that reclassify transactions deposits (demand deposits and other checkable deposits) as savings deposits overnight, thereby reducing banks' required reserves; see Anderson and Rasche (2001) and

www.stls.frb.org/research/swdata.html. For analytical purposes, MZM largely replaces M1. The **Discount Rate** and **Expected Federal Funds Rate** shown in the chart **Reserve Market Rates** are plotted as of the date of the change, while the **Effective Federal Funds Rate** is plotted as of the end of the month. Interest rates in the table are monthly averages from the Board of Governors H.15 Statistical Release. The **Treasury Yield Curve** shows constant maturity yields calculated by the U.S. Treasury Department for securities with 3 months and 1, 2, 3, 5, 7, 10, 20, and 30 years to maturity. Daily data and descriptions are available at www.stls.frb.org/fred/data/wkly.html. See also *Federal Reserve Bulletin*, table 1.35.

Page 5: **Total Checkable Deposits** is the sum of demand and other checkable deposits. **Total Savings Deposits** is the sum of money market deposit accounts and passbook and statement savings. **Time Deposits** have a minimum initial maturity of 7 days. **Large Time Deposits** are deposits of \$100,000 or more. **Retail** and **Institutional Money Market Mutual Funds** are as included in M2 and the non-M2 component of M3, respectively.

Page 7: **Excess Reserves plus RCB (Required Clearing Balance) Contracts** equals the amount of deposits at Federal Reserve Banks held by depository institutions but not applied to satisfy statutory reserve requirements. (This measure excludes the vault cash held by depository institutions that is not applied to satisfy statutory reserve requirements.) **Consumer Credit** includes most short- and intermediate-term credit extended to individuals. See *Federal Reserve Bulletin*, table 1.55.

Page 8: **Inflation Expectations** measures include the quarterly Federal Reserve Bank of Philadelphia *Survey of Professional Forecasters*, the monthly University of Michigan Survey Research Center's *Surveys of Consumers*, and the annual Federal Open Market Committee range as reported to the Congress in the February Humphrey-Hawkins Act testimony each year. Beginning February 2000, the FOMC began using the Personal Consumption Expenditures (PCE) price index to report its inflation range and therefore is not shown on this graph. **CPI Inflation** is the percentage change from a year ago in the CPI for all urban consumers. **Real Interest Rates** are ex post measures, equal to nominal rates minus CPI inflation.

Page 9: **FOMC Expected Federal Funds Rate** is the level (or midpoint of the range, if applicable) of the federal funds rate that the staff of the Federal Open Market Committee expected to be consistent with the desired degree of pressure on bank reserve positions.

Page 10: **Federal Funds Rate and Inflation Targets** shows the observed federal funds rate, quarterly, and the level of the funds rate implied by applying Taylor's (1993) equation

$$f_t^* = 2.5 + \pi_{t-1} + (\pi_{t-1} - \bar{\pi})/2 + 100 \times (y_{t-1} - y_{t-1}^p)/2$$

to five alternative target inflation rates, $\bar{\pi}^* = 0, 1, 2, 3, 4$ percent, where f_t^* is the implied federal funds rate, π_{t-1} is the previous period's inflation rate (PCE) measured on a year-over-year basis, y_{t-1} is the log of the previous period's level of real GDP, and y_{t-1}^p is the log of an estimate of the previous period's level of potential output. **Potential Real GDP** is as estimated by the Congressional Budget Office.

Monetary Base Growth and Inflation Targets shows the quarterly growth of the adjusted monetary base (modified to include an estimate of the effect of sweep programs) implied by applying McCallum's (1988, 1993) equation

$$\Delta MB_t^* = \bar{\pi}^* + (10\text{-year moving average growth of real GDP}) \\ - (4\text{-year moving average of base velocity growth})$$

to five alternative target inflation rates, $\bar{\pi}^* = 0, 1, 2, 3, 4$ percent, where ΔMB_t^* is the implied growth rate of the adjusted monetary base. The 10-year moving average growth of real GDP for a quarter "t" is calculated as the average quarterly growth during the previous 40 quarters, at an annual rate, by the formula $((y_t - y_{t-40})/40) \times 4 \times 100$, where y_t is the log of real GDP. The four-year moving average of base velocity growth is calculated similarly. To adjust the monetary base for the effect of retail-deposit sweep programs, we add to the monetary base an amount equal to 10 percent of the total amount swept, as estimated by the Federal Reserve Board staff. These estimates are imprecise, at best. Sweep program data are available at www.stls.frb.org/research/swdata.html.

Page 11: **Implied One-Year Forward Rates** are calculated by this Bank from Treasury constant maturity yields. Yields to maturity, $R(m)$, for securities with $m = 1, \dots, 30$ years to maturity are obtained by linear interpolation between reported yields. These yields are smoothed by fitting the regression suggested by Nelson and Siegel (1987),

$$R(m) = a_0 + (a_1 + a_2)(1 - e^{-m/50})/(m/50) - a_2 \times e^{-m/50},$$

and forward rates are calculated from these smoothed yields using equation (a) in table 13.1 of Shiller (1990),

$$f(m) = [D(m)R(m) - D(m-1)] / [D(m) - D(m-1)],$$

where duration is approximated as $D(m) = (1 - e^{-R(m) \times m}) / R(m)$. These rates are linear approximations to the true instantaneous forward rates; see Shiller (1990). For a discussion of the use of forward rates as indicators of inflation expectations, see Sharpe (1997). **Rates on 3-Month Eurodollar Futures and Rates on Selected Fed Funds Futures Contracts** each trace through time the yield on three specific contracts. **Implied Yields on Fed Funds Futures** displays a single day's snapshot of yields for contracts expiring in the months shown on the horizontal axis. **Inflation-Protected Treasury Yields** are yields on the most recently issued inflation-protected securities of 10- and 30-year original maturity. **Inflation-Protected Treasury Yield Spreads** equal the differences between the Treasury constant maturity yields and yields on the most recently issued inflation-protected securities of similar original maturity. **Inflation-Indexed Bonds** are, for Canada, the 31-year bond with a maturity date of 12/01/2026; for the U.K., the 37.5-year bond with a maturity date of 07/17/2024 and the 12.1-year bond with a maturity date of 10/21/2004; and, for the U.S., the 30-year bond with a maturity date of 04/15/2028 and the 10-year bond with a maturity date of 01/15/2007.

Page 12: **Velocity** (for MZM and M2) equals the ratio of GDP, measured in current dollars, to the level of the monetary aggregate. **MZM** and **M2 Own Rates** are weighted averages of the rates received by households and firms on the assets included in the aggregates. Two alternative opportunity costs are shown, one relative to the 3-month Treasury constant-maturity yield, the other to the 5-year constant-maturity yield.

Page 13: **Real Gross Domestic Product** is GDP as measured in chained 1996 dollars. The **Gross Domestic Product Price Index** is the implicit price deflator for GDP, which is defined by the Bureau of Economic Analysis, U.S. Department of Commerce, as the ratio of GDP measured in current dollars to GDP measured in chained 1996 dollars.

Page 14: **Investment Securities** are all securities held by commercial banks in both investment and trading accounts.

Sources

Bank of Canada

Canadian inflation-linked bond yields.

Bank of England

U.K. inflation-linked bond yields.

Board of Governors of the Federal Reserve System

Monetary aggregates and components, nonfinancial debt: H.6 release. Bank credit and components: H.8 release. Consumer credit: G.19 release. Required reserves, excess reserves, clearing balance contracts, and discount window borrowing: H.4.1 and H.3 releases. Interest rates: H.15 release. Nonfinancial commercial paper: Board of Governors web site. M2 own rate.

Bureau of Economic Analysis

Gross domestic product.

Bureau of Labor Statistics

Consumer price index.

Federal Reserve Bank of Philadelphia

Survey of Professional Forecasters inflation expectations.

Federal Reserve Bank of St. Louis

Adjusted monetary base and adjusted total reserves, monetary services index, MZM own rate, one-year forward rates.

Organization for Economic Cooperation and Development
International interest and inflation rates.

University of Michigan Survey Research Center

Median expected price change.

Congressional Budget Office

Potential real GDP.

Dow Jones and Co. (Wall Street Journal)

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Stock price-earnings ratio, stock price composite index.

U.S. Department of the Treasury

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Note: Articles from this Bank's *Review* are available on the Internet at www.stls.frb.org/research/index.html.